Zhang et al. 2021 — Renal Cell Carcinoma (Kidney)

CCCA embryonic-program cell-state analysis · autonomous re-run

GO / NO-GO GATES

GATE 1 · Integration + continuity — GO

7 samples (>=369-cell threshold) on ONE connected component; clear-cell samples intermix (per-neighbourhood iLISI 3.0/7 at theta=5); the chromophobe histology forms a resolved but ADJOINING region of the same continuous manifold. Cell cycle negligible (3%). No islands.

GATE 2 · Embryonic module geography — GO

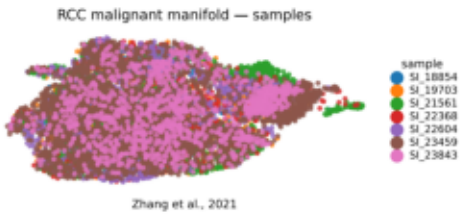
Non-branching case: a shared Gast/pSom gradient with Crest as an inverse TERMINAL gradient (pSom vs Crest r = -0.91; Crest-high decile at pseudotime 0.78). No Gast/pSom hub diverging into two competing arms. Programs present but not resolved into a bifurcation.

GATE 3 · Hub cross-repression (branching only) — NA

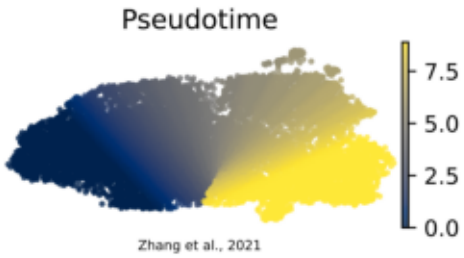
Not applicable — geometry is a single curve (non-branching), so there is no hub to test. An aggressive PPT tree over-fits the compact cloud into 33 spurious forks, not one Y.

STEP FIGURES

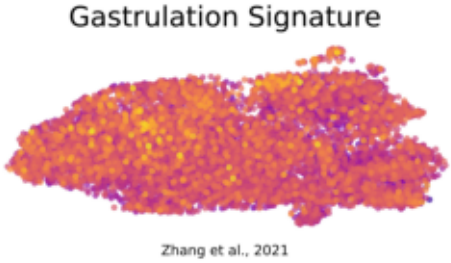
Gate 1 · Sample integration



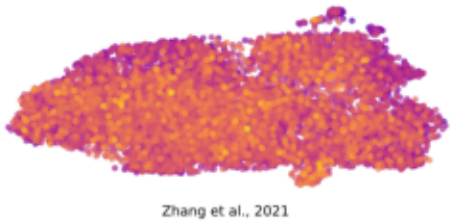
Trajectory · pseudotime (root = Gast/pSom origin)



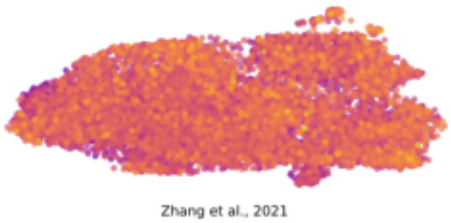
Gastrulation



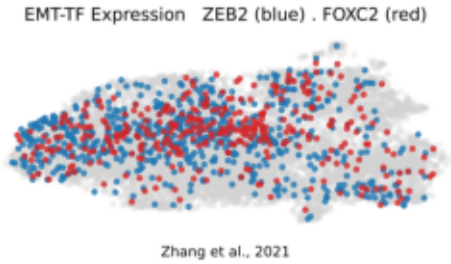
Somitogenesis



Neural Crest (terminal)



EMT-TFs · ZEB2 (blue) · FOXC2 (red)



DATASET Renal cell carcinoma (clear-cell + chromophobe) · 10x scRNA-seq · 11,218 malignant cells, 7 samples

VERDICT Pattern reproduces as the NON-BRANCHING case: a single gastrulation/somitogenesis axis terminating in a neural-crest gradient. No Gast/pSom bifurcation (Linear geometry).

KEY PARAMETERS & RATIONALE

Min-cell threshold	≥369 (geom-mean of 710→192 gap, 3.7×); drops 1 small sample → 7 kept
Integration / PCs	Harmony θ=5, n_pcs=20 (no sc.pp.scale); iLISI-selected, per-neighbourhood
Geometry	Linear — scFates principal curve (2 tips, 0 forks, fold ratio 2.71)
Root	Curve tip with max Gast+pSom, min Crest (developmental origin)
Module means	Gast −0.034 · pSom +0.085 · Crest −0.060 (score_genes, union pool)
EMT-TFs shown	ZEB2 (23.8%) + FOXC2 (12.2%) — best-detected, distinct Zeb/Foxc families